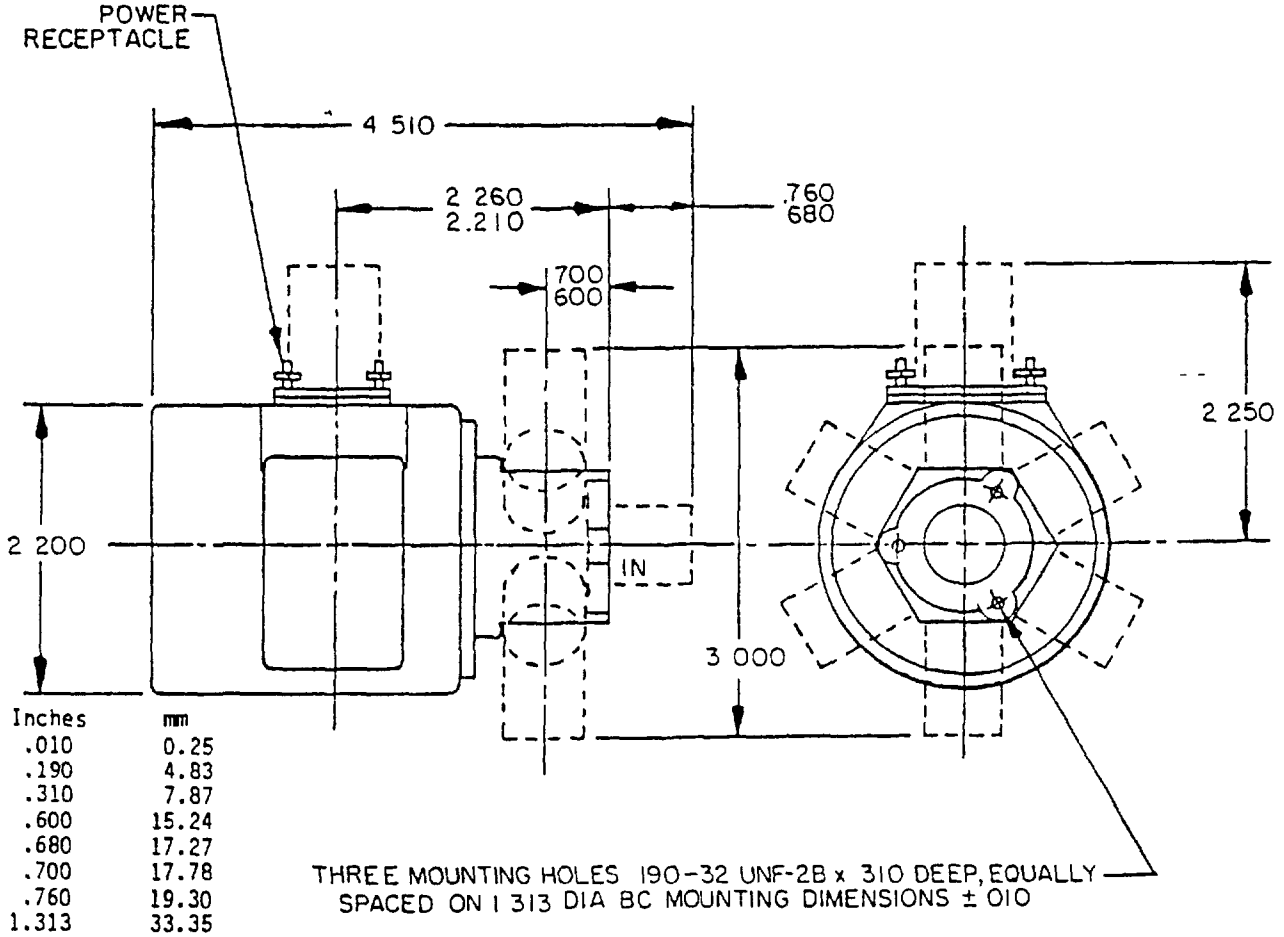
MILITARY SPECIFICATION SHEET

SWITCHES, RADIO FREQUENCY
Ⓒ TRANSMISSION LINE (COAXIAL) (MANUALLY AND SOLENOID OPERATED)
CLASSES 1 (TYPE N CONNECTORS) AND 7 (TYPE BNC CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-3928.



Inches	mm
.010	0.25
.190	4.83
.310	7.87
.600	15.24
.680	17.27
.700	17.78
.760	19.30
1.313	33.35
2.200	55.88
2.210	56.13
2.250	57.15
2.260	57.40
3.000	76.20
4.510	114.55

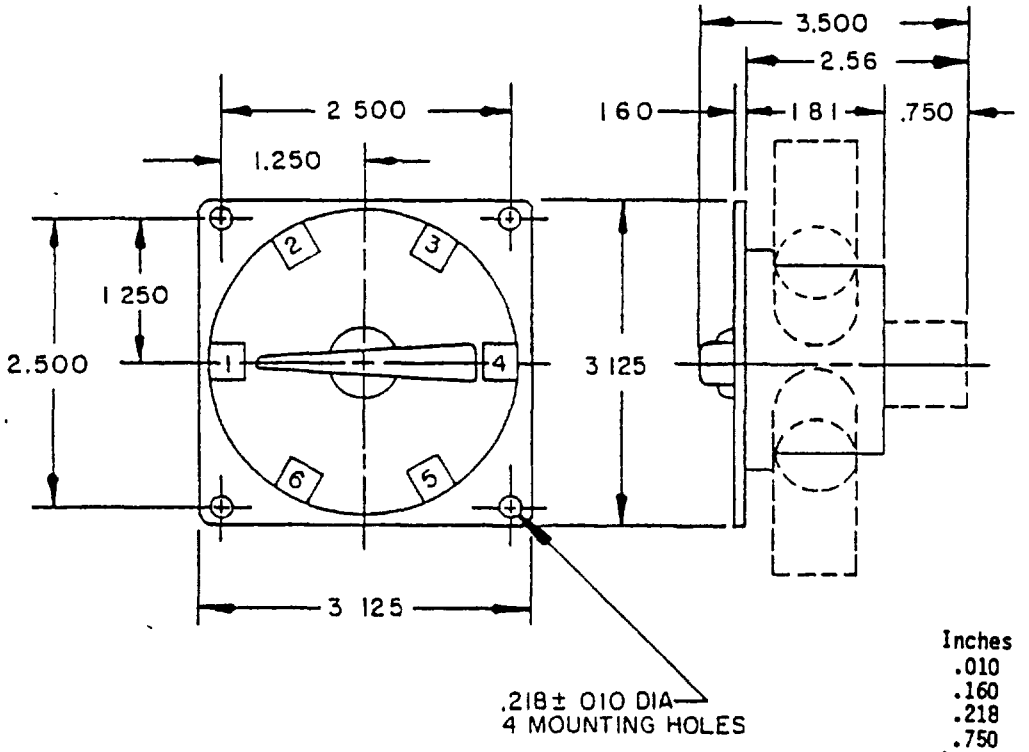
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals.

Ⓒ FIGURE 1. Dimensions and configuration for part number M3928/7-11

Ⓒ denotes changes

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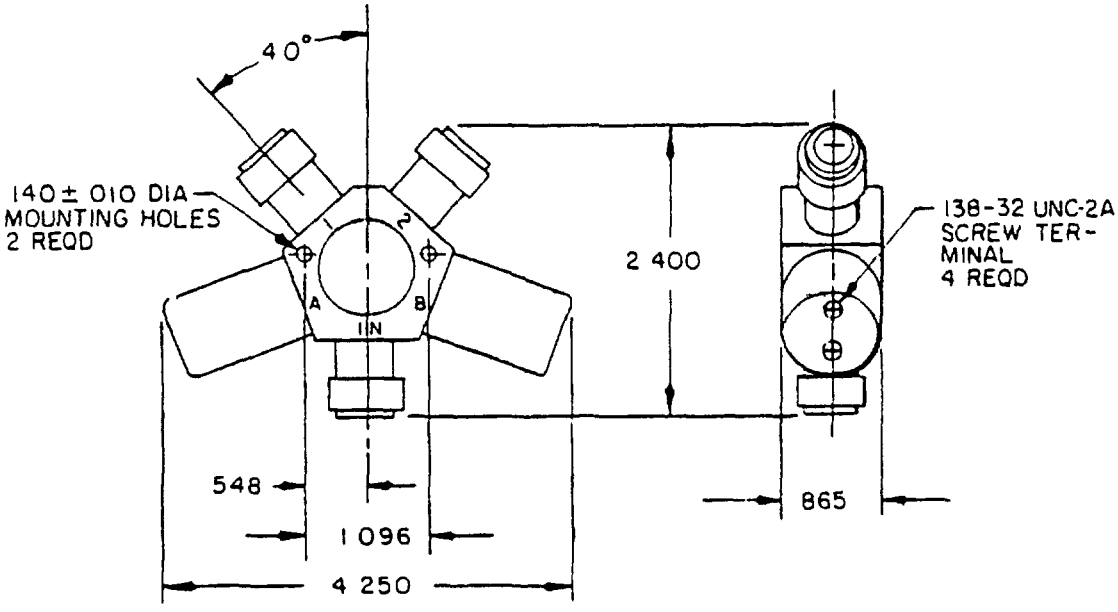


Inches	mm
.010	0.25
.160	4.06
.218	5.54
.750	19.05
1.250	31.75
1.81	46.0
2.500	63.50
2.56	65.0
3.125	79.38
3.500	88.90

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.76 mm) for two place decimals.

FIGURE 2. Dimensions and configuration for part numbers M3928/7-02, M3928/7-06, M3928/7-12, M3928/7-13, and M3928/7-22.

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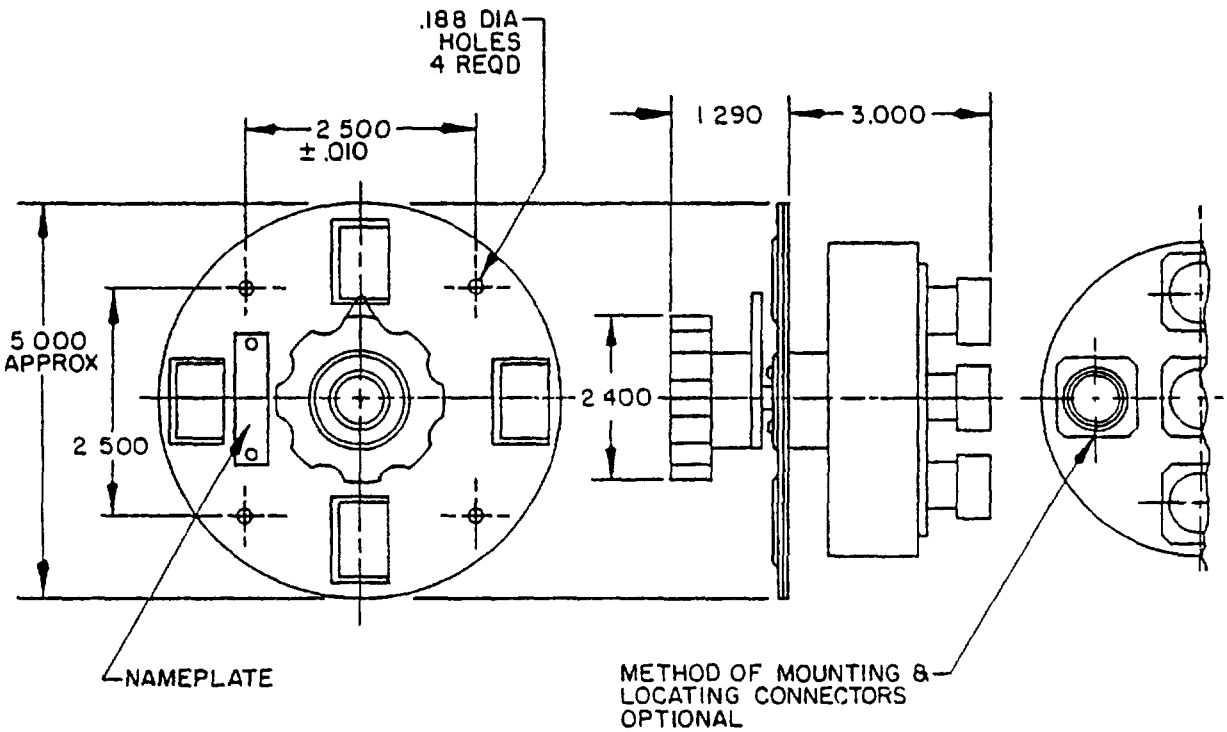
POSITION 1 – NORMALLY OPEN
POSITION 2 – NORMALLY CLOSED
POSITION IN – COMMON

Inches	mm
.010	0.25
.138	3.51
.140	3.56
.548	13.92
.865	21.97
1.096	27.84
2.400	60.96
4.250	107.95

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.76 mm) for two place decimals.

FIGURE 3. Dimensions and configuration for part numbers M3928/7-09, M3928/7-10, M3928/7-17, M3928/7-18, M3928/7-19, and M3928/7-20.

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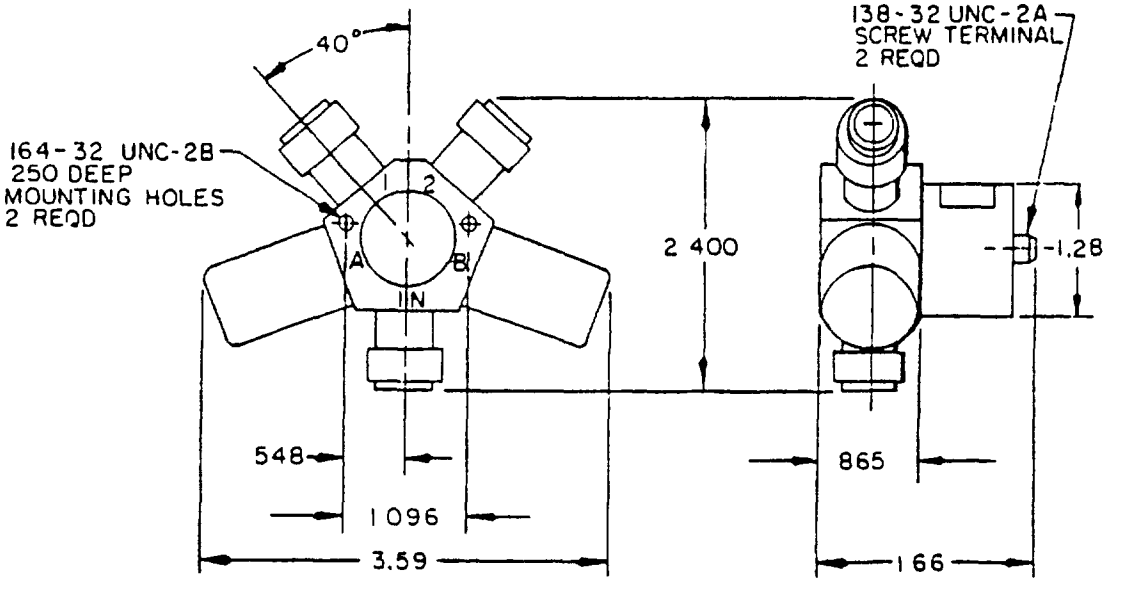


Inches	mm
.010	0.25
.188	4.78
1.290	32.77
2.400	60.96
2.500	60.96
3.000	76.20
5.000	127.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals.

FIGURE 4. Dimensions and configuration for part numbers M3928/7-14, M3928/7-15, and M3928/7-16.

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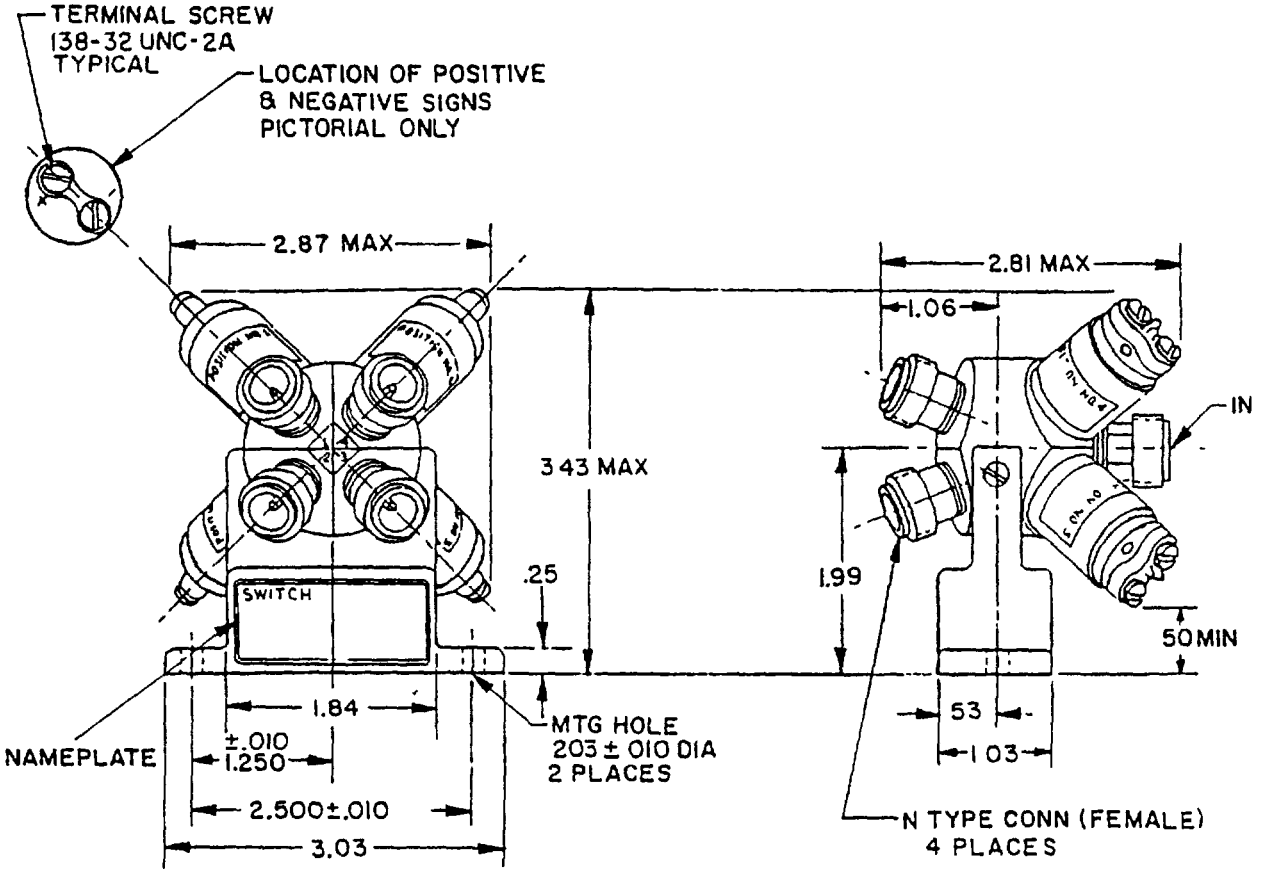
POSITION 1 – NORMALLY OPEN
POSITION 2 – NORMALLY CLOSED
POSITION IN – COMMON

Inches	mm
.138	3.51
.164	4.17
.250	6.35
.548	13.92
.865	21.97
1.096	27.84
1.28	32.5
1.66	42.2
2.400	60.96
3.59	91.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.76 mm) for two place decimals.

FIGURE 5 Dimensions and configuration for part number M3928/7-23

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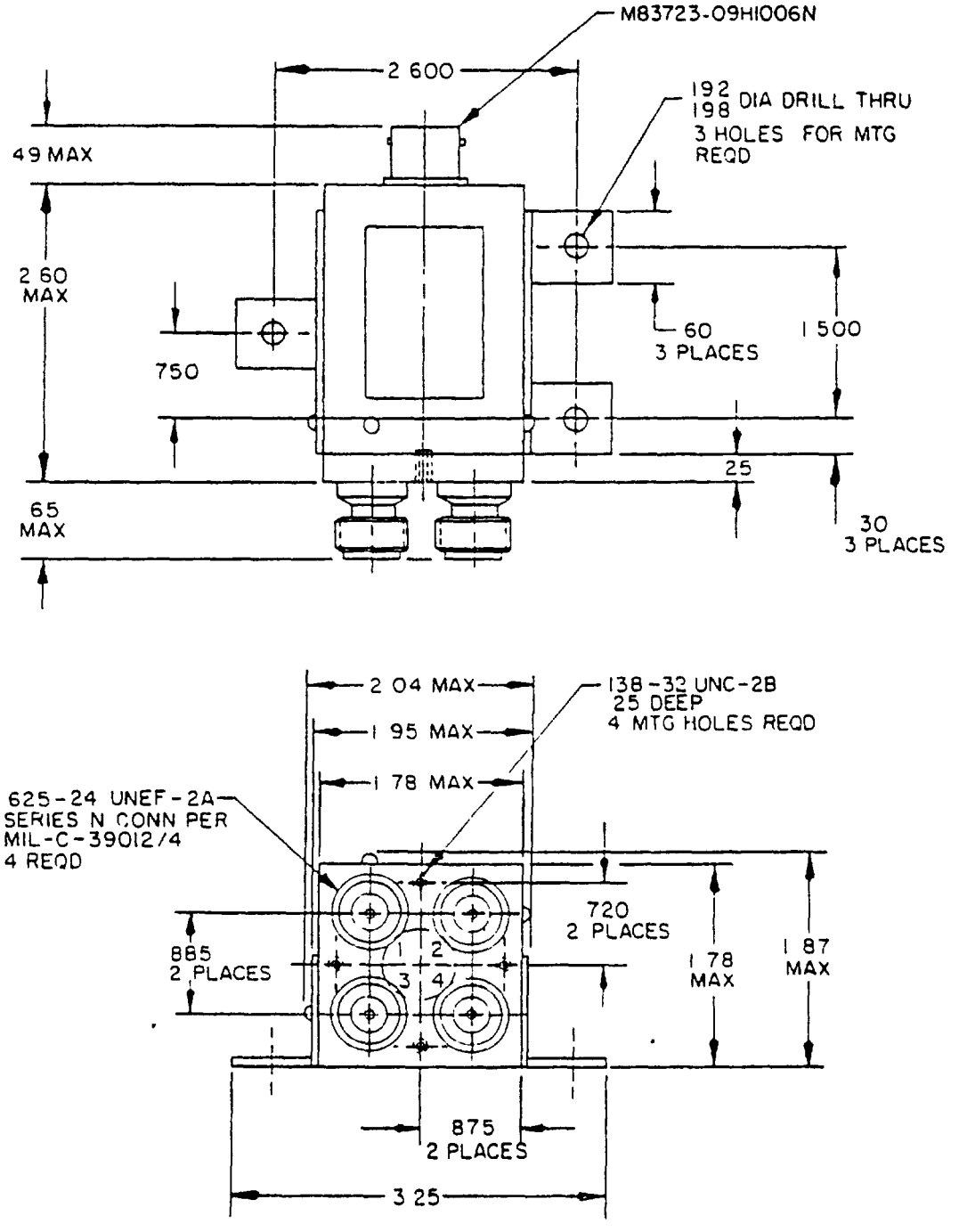


Inches	mm
.010	0.25
.138	3.51
.203	5.16
.25	6.4
.50	12.7
.53	13.5
1.03	26.2
1.06	26.9
1.250	31.75
1.87	47.5
1.99	50.6
2.500	63.50
2.81	71.4
2.87	72.9
3.03	77.0
3.43	87.1

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.76 mm) for two place decimals.

FIGURE 6. Dimensions and configuration for part number M3928/7-24.

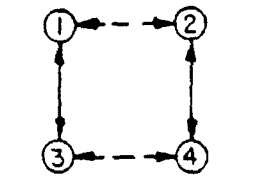
Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:15Z
Check the source to verify that this is the current version before use.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.76 mm) for two place decimals.

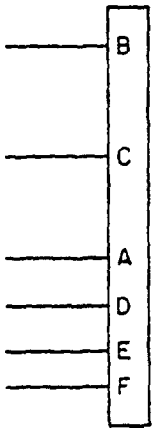
FIGURE 7. Dimensions and configuration for part numbers M3928/7-25 and M3928/7-26

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POSN 1 ⟷ 1-3,2-4
POSN 2 ⟷ 1-2,3-4

R F CIRCUIT



POWER AND INDICATOR CIRCUIT

Inches	mm	Inches	mm
.138	3.51	.875	22.23
.192	4.88	.885	22.48
.198	5.03	1.500	38.10
.25	6.4	1.78	45.2
.30	7.6	1.87	47.5
.49	12.5	1.95	49.5
.60	15.2	2.04	51.8
.65	16.5	2.600	66.04
.720	18.29	3.25	82.6
.750	19.05		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.76 mm) for two place decimals.
4. Remove all burrs and sharp edges .015 (0.38 mm) maximum.
5. All machined surfaces shall be in accordance with ANSI B46.1.

FIGURE 7. Dimensions and configuration for part numbers M3928/7-25 and M3928/7-26 - Continued.

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© TABLE I Electrical and performance characteristics

Part no M3928/7-	Figure no	Housing	Manual or remote	Solenoid or motor	Fail safe or latching	RF conn type	Frequency range dc to GHz	VSWR	Insertion loss (dB)	Isolation (dB)	Con-fig	Position indicating circuit (and current rating)	Life cycles x 1000	Nominal oper voltage	Oper current (A)	Pickup voltage	Dropout voltage	Switch time (ms)	Power connector	Weight (lbs)
								max	max	min					max					max
02N,S	2	I	M	---	- -	N	10	1 5	0 52	25	1P2T	None	25	---	---	--	---	---	--	1
06N S	2	I	M	---	---	N	10	1 5	0 52	25	1P6T	None	25	---	---	--	---	---	---	1-1/4
09N,S	3	I	R	S	F	N	10	1 5	0 52	20	1P2T	None	1000	110 dc	2	60	2	20	Screw 6-32NC-2A	1/2
10N,S	3	I	R	S	F	BNC	10	1 5	0 52	25	1P2T	None	1000	26 dc	3	18	0 5	20	Screw 6 32NC-2A	1/2
11N,S	1	I	R	S	---	N	10	1 5	0 52	25	1P3T	None	100	110 dc	3	60	2	250	MS3456-14S-2S	2
12N,S	2	I	M	---	---	N	10	1 5	0 52	25	1P3T	None	25	---	---	---	---	---	---	1
13N S	2	I	M	---	---	BNC	10	1 5	0 52	25	1P6T	None	25	---	---	---	---	---	--	1-1/4
14N,S	4	I	M	- -	---	N	10	1 5	0 52	25	1P2T	None	25	---	---	---	---	---	---	1-5/16
15N,S	4	I	M	---	---	N	10	1 5	0 52	25	1P6T	None	25	---	---	---	---	---	---	1-1/2
16N,S	4	I	M	---	---	N	10	1 5	0 52	25	1P4T	None	25	---	---	---	---	---	---	1-1/2
17N S	3	I	R	S	F	N	10	1 5	0 52	20	1P2T	None	1000	26 dc	2	18	0 5	20	Screw 6-32NC-2A	1/2
18N,S	3 1/	I	R	S	F	N	10	1 5	0 52	25	1P2T	None	1000	26 dc	2	18	0 5	20	Screw 6-32NC-2A	1/2
19N,S	3 1/	I	R	S	F	N	10	1 5	0 52	25	1P2T	None	1000	110 dc	2	60	2	20	Screw 6-32NC-2A	1/2
20N,S	3 2/	I	R	S	F	N	10	1 5	0 52	25	1P2T	None	1000	28 dc	2	18	0 5	20	Screw 6-32NC-2A	1/2
22N,S	2	I	M	--	- -	N	10	1 5	0 52	25	1P4T	None	25	---	---	---	---	--	---	1
23N,S	5	I	R	S	F	BNC	10	1 5	0 52	25	1P2T	None	1000	115 V ac 3/	2	93	5	20	Screw 6-32NC-2A	1/2
24N,S	6	I	R	S	---	N	10	1 5	0 52	25	1P4T	None	1000	28 V dc	2	18	0 5	20	Screw 6-32NC-2A	3/4
25N,S	7	I	R	S	L	N	12 4	1 5	0 5	60	TR	None	100	28 V dc	0 64	20	---	20	MSB3723-13R1006N	3/4
26N,S	7	I	R	S	L	N	12 4	1 5	0 5	60	TR	Yes 4A res 2A ind	100	28 V dc	0 64	20	---	20	MSB3723-13R1006N	3/4

1/ Both positions normally open

2/ Both positions normally closed

3/ Frequency 400 Hz

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REQUIREMENTS:

Dimensions and configurations: See figures 1 through 7.

Electrical and performance characteristics: See table I.

RF power handling capability (average): 100 watts (minimum).

RF connectors: Series N connectors shall mate with series N male connectors per MIL-C-39012/1. BNC connectors shall mate with BNC male connectors per MIL-C-39012/16.

Nominal impedance: 50 ohms.

Termination: Open.

Vibration: Method I.

Operating temperature: -55°C to +85°C.

Part number: M3928/7- (and dash number from table I).

© TABLE II. Part number to type cross-reference. 1/

Part number M3928/7-	Type SA- /U	Part number M3928/7-	Type SA- /U
02	1321	17	---
06	1325	18	---
09	1328	19	---
10	1329	20	---
11	1330	22	---
12	1331	23	---
13	1332	24	---
14	273	25	---
15	275	26	---
16	274		

1/ Part numbers M3928/7-01, M3928/7-03, M3928/7-04, M3928/7-05, M3928/7-07, M3928/7-08, and M3928/7-21 have been deleted.

NOTE: Switches covered by this specification sheet are substitutable for the manufacturer's or original equipment manufacturer's (OEM) part numbers specified in table III. This information in no way implies that the manufacturer's or OEM part number or NSN is suitable as a substitute for the military part number.

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© TABLE III Substitution information.

Part number M3928/7-	OEM or manufacturer's code	OEM or manufacturer's part number	NSN 5985
02	82152	CON2AB	00-552-9040
02	74868	300-81792-11	---
06	82152	CON6AB	00-754-9860
06	74868	300-81792-19	---
09	82152	11600	---
10	82152	13300	---
11	74868	300-82002-1146	---
12	74868	300-81792-14	---
13	74868	300-81792-29	---
14	74868	300-82350-211	---
15	74868	300-82350-217	00-884-2679
16	74868	300-82350-219	---
17	82152	11300	00-504-8506
18	82152	11100	00-557-5208
19	82152	11400	---
20	82512	11200	00-557-5721
22	82512	CON4AB	---
24	82512	14100	00-501-1886
25	82512	300C00100	---
26	82512	300C00200	00-241-3503

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 99
DLA - ES

User activities:
Army - AV
Navy - AS, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent.
DLA - ES

(Project 5985-1016-2)

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